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Social media: Benefit vs harm

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Introduction

Social media is a relatively inexpensive and readily available electronic resource that allows anyone to publish and access content and cooperate on a collective effort, becoming an essential part of modern culture, affecting all races, sexes, socioeconomic backgrounds, and people all over the world. It is becoming apparent that social networking websites have become an integral part of people's lives. As a result, there is a need for people to adjust their lifestyles as emerging technologies evolve. These platforms have been a part of people's daily lives, with many applications ranging from education to corporate contact, marketing, healthcare, and even dating.

Social media is a form of web-based data communication where users may hold discussions, exchange information, and generate web material. Individually, allowing people to connect with our friends and families, learn new knowledge, cultivate our passions, and be entertained (Akram & Kumar, 2017). On a professional level, it can extend or deepen expertise in a specific field and develop a professional network by engaging with other professionals within the industry (Ji et al., 2010). Social media is a new concept with tremendous growth potential. Similarly, people no longer need to depend on the newspapers or television to get their regular news consumption; it can all be accessed on a social networking site (Bolier et al., 2013). As a result, studying all the possible dangers and advantages is critical because it will offer a well-rounded solution to improving our present position in an environment where social networking can cause harms such as addiction or cyberbullying.

The objective of this study is to focus on literature from both small and large studies examining the harms and advantages of the online environment, to identify the line at which point casual social media use may become dangerous and to highlight possible avoidance techniques to enable the virtual world to become less harmful.



Figure 1. Illustrating the aims and objectives of the report

Social media use

In the 2006 Parents and Teens survey, a random sample of 935 participants demonstrated that 55% of youths used SNSs (social networking sites) in that year. Furthermore, they also investigated that 91% used social media to stay in touch with friends and 49% to make new friends (more common in boys than girls) (Lenhart & Madden, 2007). The most common activities on SNSs were reading, responding to comments (60%) and posting comments, links, responding to messages (14%), and browsing profiles (13%) (Pempek et al., 2009).

In another study discussing how people interact with SNSs using psychophysiological measures (skin conductance and facial electromyography), media scanning (extracting material from profiles) was found to be more rewarding than social surfing (passive reading of news feeds) (Bolls et al., 2009). Consequently, the results indicate that goal-directed social seeking can activate the appetitive system rather than the aversive system due to a genetic defect in the addict's neurochemical reward system (Uylings et al., 2000).

According to the gratification hypothesis, media is seen as a goal-directed means of gratification and need fulfilment and is comparable to addiction. People with higher social identity (i.e., collective solidarity/conformity), altruism, and telepresence are more likely to use SNSs because they perceive support for social network participation (Kwon & Wen, 2010). A survey of 170 university students found that social influences were more important motivators for SNS use than human factors (Kim et al., 2010).

Another study found that collective self-esteem and group identification positively correlated with peer group communication via SNSs, suggesting that people being part of the same virtual realm, the support of group values, the preservation of interpersonal interconnectivity, and social change were some of the reasons for SNS usage (Cheung et al., 2011).

The participating university student sample endorsed the following motivations:

- To keep in contact with friends they do not often see. (81%)
- To conform to other people/friends who have them be part of a group. (61%)
- To keep in contact with relatives and family. (48%)
- To make plans with friends they often see. (35%)

Overall, it appears that the advantages of bridging social capital created by engagement in SNSs were especially beneficial for people with low self-esteem (Kujath, 2011). Furthermore, due to many possible low social links between participants, which are allowed by digital media's structural features, SNSs are thought to increase a potential

networks' size (Donath & Boyd, 2004). Specifically, maintaining social capital via participation in SNSs is more beneficial to students regarding potential employment opportunities and sustaining ties with old friends. Thus, one explanation why people with low self-esteem are attracted to using SNSs in a potentially excessive way is the ease of creating and retaining bridging social capital and, in combination with lower self-esteem, has been demonstrated to be related to Internet addiction (Armstrong et al., 2000).

A study on SNS usage between different cultures (the USA vs Korea vs China) has demonstrated that using different Facebook functions was associated with creating and maintaining either bridging or bonding social capital. For example, people in the US used SNSs for communication, whereas Koreans and Chinese used SNSs for expert search (i.e., searching for professionals online) and 'Connection' (i.e., maintaining offline relationships) (Ji et al., 2010).

A student survey study of 387 participants found the predictive factors were: playfulness (i.e., enjoyment and pleasure), the critical mass of the users who endorsed the technology, trust in the site, perceived ease of use, and perceived usefulness (Sledgianowski & Kulviwat, 2009).

The studies contained small convenience samples, teenagers, or university students as participants, therefore severely limiting the generalisability of findings.

Using the gratification theory demonstrates how popular social media is in today's world and how social media as a whole is tailored to our social needs. It is especially important now to stay in touch online for the evolution of friendships, not just one-on-one friendships, but also as a group and for personal and business purposes worldwide. As a result, 81% of social media users use it to communicate with current friends they do not see very much. SNSs also provide introverted individuals with the opportunity for better self-expression.

Social media and neuronal systems

According to the study by Meshi et al. (2015), social media serves as an online platform for meeting basic social needs via five critical strategies. These five behaviours: Users broadcasting information, receiving feedback, observing, providing feedback and engaging in social comparison, depend on three domains. The three domains are cognitive processing (mentalising), self-referential cognition, and social reward processing. According to the mentalising network, social media allows one to consider other people's emotional states/motivations. The self-referential cognition network provides and embraces input expressing self-evaluation and social distinction. Finally, the incentive network implies that social media offers a consistent supply of social incentives and reinforcement in positive feedback or suggestions, resulting in credibility enhancement. Because of the minimalistic cue of social performance, both stimuli stimulate the reward system (Meshi et al., 2015).

Social rewards activate a network of brain regions, including the ventromedial prefrontal cortex (VMPFC), the ventral striatum, and the ventral tegmental lobe (Fareri & Delgado, 2014). This knowledge has been extended to practical values by connecting social media stimuli and brain activation regions to observe the social values people have outside of social media. Exchanging information with others stimulates the VMPFC and ventral striatum. Receiving positive social reinforcement (e.g., signals that people understand you, agree with you or like you) and providing others with the same social incentives (e.g., offering a 'like') is a form of prosocial behaviour triggering the reward mechanism (such as donating to charity). Reading other people's posts offers the same reward activity as acquiring information and is followed by activity in the ventral striatum. According to analysis, the ventral striatum also underpins social comparison, which suggests that behaviour in this area shows the comparison between one's reward and that of another, which could suggest an explanation to the form of jealousy people experience whilst on social media (Fareri & Delgado, 2014).

Social media's main advantages are that it allows communication not constrained by spatial and temporal distance. However, people are likely to be deprived of social cues such as close interaction, e.g., seeing micro-expressions and feeling physical contact. It was observed that individuals with more extensive online social networks have a larger amygdala than individuals with smaller social networks, which suggests a physiological reason for the attraction of humans to the use of social media (Meshi et al., 2015).

A study examining the relationship between online social media use and the neural processing of offline credibility benefits showed that the nucleus accumbent reaction to reputation enhancement predicted participants' social media use intensity. These findings support these processes' role in the sustainability of social networks and add to our

knowledge of the factors that influence social behaviour. More specifically, they demonstrate the feasibility of studying social neurocognitive processes using metrics created by social media use (Meshi et al., 2013).

Collectively, these findings exhibit how social media is tailored and adapted to the human evolutionary urge to socialise, learn, and communicate with groups of people. Thus, neuropsychology data explains how people are programmed to communicate and how social media networks are designed to satisfy and even exceed those needs, providing the backbone to both potential harms research and social networks' overall usefulness and functionality.

Social media addiction

In recent years, with the development of information technology, especially the rapid spread of internet-based social media (e.g., Facebook, WeChat, or Instagram), the ways of interpersonal communication have dramatically changed (Anderson & Smith, 2018). The ubiquitous social media platforms and easy access to the internet give rise to the potential for social media addiction, i.e., the irrational and excessive use of social media to the extent that it interferes with other daily aspects of life (Griffiths, 2009). There is considerable evidence demonstrating a connection between social media addiction and a range of emotional, interpersonal, health, and performance problems (Griffiths, 2000).

Social media addiction is a type of internet addiction in which people feel compelled to use social media in excessive amounts (Griffiths, 2000). Individuals addicted to social media are often unnecessarily worried and are motivated by an uncontrollable compulsion to log in to utilise it (Andreassen & Pallesen, 2014). According to studies, mood, cognition, physical and emotional reactions, and interpersonal and psychological problems are all social media addiction symptoms (Błachnio et al., 2017). According to reports, approximately 12% of social networking sites users suffer from social media addiction (Wu et al., 2013).

Recent literature on social media usage and mental wellbeing has shown that long-term use of social media platforms like Facebook is positively correlated with mental health problems like stress, anxiety, and depression while negatively associated with long-term wellbeing (Eraslan-Capan, 2015). Time spent on social media, for example, was linked to depressive symptoms in high school students in Central Serbia (Pantic et al., 2012) and young adults in the United States (Lin et al., 2016).

The argument about whether so-called internet addiction should be accepted as a disease in the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition is ongoing. The correlation between “addiction” and various compulsive or impulsive activities is often a source of misunderstanding. Some psychologists claim that internet addiction has the same traits as many other substance use disorders, such as heavy use, withdrawal effects, tolerance, and harmful consequences; however, there is no physiological research to support these arguments. Thus, it is uncertain if internet addiction is a symptom of an underlying condition or a different illness category. In addition, the recurrent emergence of internet addiction in the sense of a wide range of comorbidities poses difficult causality questions (Pies, 2009).

A more generic model of what qualifies as “disease” is required and as a particular disease to make nosological conclusions on internet addiction. It remains premature to recognise internet addiction as a distinct disease category, based on a model addressing intrinsic distress and incapacity and evidence on course, prognosis, temporal

consistency, and recovery response. However, recent evidence shows that certain people addicted to the internet are at severe risk and need immediate support. To overcome these disputes, closely conducted studies are needed (Pies, 2009).

Social media and cell phone addiction

Addiction can be anything if there is a possibility of negative consequences and physical and psychological reinforcements in a specific context – cell phone and social media addiction would be classed as a behavioural addiction. In the case of behaviour, it is difficult to determine whether problems result from problematic behaviour, personality traits, or psychiatric comorbidities (De-Sola Gutiérrez et al., 2016). Young, (1999) studied five different forms of addictive behaviour on the internet: firstly, the computer itself, the act of searching for information, interaction compulsions, such as contact with the web through online games, shopping, etc., cyber sexuality, and cyber contacts (Young, 1999).

In general, it is noted that addiction entails abuse without control, alterations in mood, tolerance, abstinence, personal harm or conflicts in the environment, and a tendency to relapse (Griffiths, 1996). The Cell Age Telephone initiation is getting younger as about 30% of 10-year-old Spanish children have a mobile phone, 70% of 12-year-olds and 83% have a cell phone by the age of 14, causing behavioural problems and disorders in teens.

As in general, the youngest group, mainly adolescents, are the most highly affected by and at risk for substance and behavioural addictions (Protégeles, 2014). Furthermore, this has become more evident within the communications media, inspiring new pathologies, such as “Nomophobia” (No-Mobile-Phobia), “FOMO” (Fear Of Missing Out), the fear of being deprived of a cell phone, disconnected or off the internet. It is further extending to false sensations such as “Textaphrenia” and “Ringxiety” – which is thinking that one has received a text message or a phone call when they have not, causing them to be continually checking the device (Taneja, 2014). Moreover, “Textiety” – the anxiety of receiving and responding immediately to text messages. Compared to and corroborated by the diagnostic criteria of the DSM, problematic manifestations have suggested that when in unsafe circumstances or forbidden contexts, problematic and deliberate use is a sign of an arising addiction (Billieux, 2008).

In behavioural terms, these problematic behaviours have been highlighted, compared to and commonly validated by the diagnostic criteria of the DSM: Social and familial conflicts and confrontations and the loss of interest in other activities (Griffiths, 2000), harm repeated physical, mental, social, work, or familial interruptions, preferring the cell phone to personal contact (Chesley, 2005), other addiction-like behaviours such as excessive use, urgency, tolerance, dependence, difficulty controlling, craving, increasing use to achieve satisfaction or relaxation, or counteracting a dysphoric mood (Kwon et al., 2013), anxiety and loneliness when unable to send a message or receive immediate response stress (Ha et al., 2008) and mood changes due to the need to respond immediately to messages (Sansone & Sansone, 2013).

Sociodemographic differences in cell phone dependency

Age

The data suggest that the average time spent on mobile phones declines with age, with people younger than 20 years of age showing the highest hours, mainly teens under 14 (Lopez-Fernandez et al., 2014). This phenomenon has been suggested to be due to decreased self-control observed at that age, due to underdeveloped structures in the brain which affect self-control (Bianchi & Phillips, 2005). The most frequent use of the mobile phone is in the applications which allow text messaging (Lu et al., 2011).

Cell phones have been observed to be so vital in an adolescent's life that between the ages of 13-14, one out of three never turns off their phone (Protégeles, 2014). Unfortunately, this has been found to link with other problematic addictions in the future, especially when the first phone was obtained before the age of 13 years (Sahin et al., 2013).

Gender

Most research suggests that women have higher levels of dependency and problematic usage than men (Jenaro et al., 2007). Where usually, female cell phone usage is connected to sociability (Bianchi & Phillips, 2005).

There were differences observed between female and male usage time rather than utilisation. Females have been observed to spend more time on applications that increase intimacy with other people, orientated towards intense and close relationships. In comparison, males tend to spend more time on applications for practical and instrumental reasons (Roberts et al., 2014).

The formation and management of connections, interpersonal relationships, indirect communication, and texting are the most frequently used applications for both sexes. However, there was a difference observed in cell phone and internet usage. Whilst females use cell phones as a means of social contact; they have less frequent internet usage. Hence, problematic internet behaviour is more frequently observed in males (Ruiz-Olivares et al., 2010).

Education, Cultural and Economic

Effects of culture and economic status on the dependence on social media were observed in students. It was found that students from higher cultural and economic levels showed higher levels of dependence. This was suggested to be due to more frequent isolation due to studying, where a cell phone is a tool for contact (Amidi Mazaheri & Karbasi, 2014). A significant association was observed between parents' abuse and dependence on cell

phones and children's addiction to the internet and other technologies. The correlation has been suggested to be due to emotional abandonment (Zhou et al., 2014).

Personality and Psychological Variables

The studies regarding problematic cell phone and internet usage focus on the five-factor model (FFM) of personality, self-esteem, self-concept, self-identity and impulsivity.

Personality characteristic studies have observed that extroverts use social media networks to form new and improve old relationships. In contrast, introverts use social networks to compensate for human interaction. Both introverts and extroverts have the potential to become dependent. However, it has been found that extroverts with low scores in conscientiousness and introverts with high scores in neuroticism and narcissism have the highest potential to become addicted (Kuss & Griffiths, 2011). Neuroticism was a significant factor as it results in high levels of anxiety and insecurity (Correa et al., 2010).

Problems derived from cell phone abuse

The research focuses on sleep disruption and its coexistence with the use of substances such as alcohol and tobacco and with symptomatology and psychiatric comorbidities, particularly anxiety, stress, and depression, concerning the psychological problems resulting from cell phone misuse.

It was found that problematic student cell phone usage, especially in females, correlated with high levels of both anxiety and insomnia (Jenaro et al., 2007). A link was observed between the number of phone calls and messages with sleep difficulties and a tendency to use cell phones at night (Thomée et al., 2007). Cell phone abuse has been linked with personal stress as it maintains a constant state of alertness hence interferes with sleep (Sansone & Sansone, 2013). Similarly, with social networks, excessive Facebook usage has been associated with interruption of sleep during the week and decreasing the number of hours slept (Andreassen et al., 2012).

Overall, it is evident how social media can easily and quickly grasp people across all ages, genders, socioeconomic groups and cultures. Furthermore, it also demonstrates how they can become dependent and what effects it can cause (as observed in figure 2).



Figure 2. Taxonomy of online harm, illustrating how, for most people, it is a beneficial relationship reaching symbiosis and with a clear benefit for the user. However, per so many beneficial interactions, there will be a harmful one. A person's range of environmental and genetic factors directly impacts the level of potential harm and the exposure time needed to cause harm.

Social media use in healthcare

The first research on social media impact was focused on experiences with patients and healthcare providers and the effect of social media on their wellbeing. Seven types of effects on patients were identified: improved self-management and control, enhanced psychological and subjective wellbeing, diminished subjective wellbeing, addiction to social media, loss of privacy, and being targeted for promotion. The findings revealed that the primary motive for using social media was for social support. Furthermore, it led to equal communication between the patient and healthcare professional, increased doctors' switching, harmonious relationships, and suboptimal interaction between the patient and healthcare professional (Smailhodzic et al., 2016).

The social media support was defined as "the process of interaction in relationships which is intended to improve coping, esteem, belonging, and competence through actual or perceived exchanges of psychosocial resources" (Cohen et al., 2000) as it was observed that within the social media support it covered four different areas: emotional support, esteem support, information support, and network support. Patients who used social media during their illness had more opportunities to connect with others in a similar situation, reducing loneliness (Colineau & Paris, 2010). Moreover, social media provided an opportunity for patients to discuss the sensitive characteristics of their condition as using online communities reduces the inhibitions felt during sharing experiences face-to-face (Broom, 2005). Thus, patients who used social media for network support exhibited enhanced psychological wellbeing. Despite this, it was revealed that social media could also harm a patient's wellbeing. It was observed that social media could trigger unnecessary jealousy and anxiety when patients took a 'lurker' role on social media, only using it to read stories and compare, therefore illustrating how social media's effects are highly dependent on the participant rather than the platform. It can be a haven for the patient to unload their problems and develop coping strategies, or it can be a source of anxiety and potential triggers (Smailhodzic et al., 2016).

The study provides some critical points to determine how vital social media can be and how it can produce a positively transformative medium when used correctly. However, it also demonstrates how damaging it can also be for individuals when misused. Therefore, supplying a platform where various patients can discuss and advise each other whilst minimising potential damage by preventing lurking will be an important development within the social media realm.

Social media use with students

The research looked at how social media and social media multitasking (SMM) affect the academic success of university students. The research comprised a group of 348 undergraduate students at Hong Kong's comprehensive university. The research revealed that social media use and SMM are two main variables that detrimentally affect university students' academic success, specifically student learning (Lau, 2017). Cheever et al. (2015) proposed strategies such as technology breaks, self-monitoring, teaching metacognitive skills, and encouraging technology awareness to help students navigate their use of technology, allowing social media to reach its full beneficiary potential.

Social media is said to have a detrimental academic performance effect (Jacobsen & Forste, 2011) but can also improve student learning and 'cognitive capacity' within the social context (Bell, 2011). For example, in students engaging in forums, it is assumed that since students are engaged and interested in such forums, these channels would encourage students to think more critically and objectively (Jones 2011). Furthermore, social networking facilitates collaborative learning (such as peer-to-peer and lecturer-to-many students). Since social media forums produce a long-lasting public message, discussions should be more constructive and thoughtful (Hsiao et al., 2013). As evidence suggests that more critical thinking occurs, online communities deliver more significant interaction, informative points, and personal reflection on both theory and practice (Meyer 2004). Despite this, the findings indicate that less social learning exists through online conversations compared to face-to-face discussions.

In the world of adolescents, social media and learning do not seem to go hand in hand, likely due to the apparent element of imminent distraction. However, an opportunity might exist for introverted students who lack the courage to speak up in class or for more extroverted personalities who could not do enough critical thinking before speaking up if the risk of distraction was eliminated and there is enough incentive to engage. A site with the ability to cater to both could revolutionise online education.

Social media effects on welfare

Bolier et al. (2013) produced research that looked at the implications of turning off Facebook for four weeks. A sample of 2,743 people and their willingness to agree (WTA) to get their Facebook accounts deactivated for four weeks was observed.

The sample was split into two groups: a recovery group paid \$102 and a monitoring group not paid to deactivate. It was observed that those who reduced online contact, thereby increasing offline activities such as watching TV alone and socialising with family and friends, demonstrated behaviours that: reduced both objective news and political polarisation, improved subjective wellbeing and resulted in a significant and lasting decrease in Facebook post-experiment usage. In addition, deactivation resulted in small but important wellbeing improvements such as self-reported happiness, personal satisfaction, and less depression and anxiety. The overall index of subjective wellbeing improved by 0.09 standard deviations – point of comparison used about 25–40% of the effect of psychological interventions including self-help therapy, group training, and individual therapy, as reported in a meta-analysis by (Bolier et al., 2013).

Facebook's deactivation increased subjective wellbeing, and 80% of the Recovery Community said the deactivation phase was successful for them. However, the participants could not give up Facebook unless they received reasonably significant sums of money: even after being deactivated for four weeks, which should have facilitated at least some learning or "detox" from addiction (Bolier et al., 2013). According to the results on the usage of Facebook, Facebook is a significant source of news and knowledge. In interviews, it was clear that Facebook would affect people's lives in various ways, either as a source of entertainment, a way to coordinate a charity or activist group, or a vital social lifeline for those who are otherwise alone. Despite the drawbacks of social media, it seems to satisfy essential and universal needs (Bolier et al., 2013).

It offers some critical insights into the advantages of avoiding social media and concentrating on other activities such as spending time with friends and relatives and improving general wellbeing. Despite this, the four-week cleanse demonstrates its impact on people's lives, as shown by the vast number of people returning to social media.

Social media use in dating

The dating industry has over 2 billion users, with the first domain coming into existence in 1995 and being studied systematically from 2005 (Wiederhold, 2015). There have been multiple advantages observed in using online platforms to find a prospective partner, such as broader prospects to meet potential partners, less intimidation to initiate contact, and the possibility that the questions/criteria can open someone's mind to more possibilities/criteria out of their comfort zones. However, there are also disadvantages such as objectifying nature, the possibility that the domain users may form bad decisions based on many choices, the potential to delay face-to-face contact for too long, and misinterpretation of messages (Finkel et al., 2012).

Ultimately, the dating domains should only be used as a tool to find a potential partner, as genuine relationships can only begin after face-to-face contact. Research into online dating has demonstrated that eventually, it will reside at the intersection of experiential attributes, chemistry, and gut-level evaluations. Specific emotional reactions are based on sensory experiences such as olfaction, which cannot be obtained any other way. In-person meeting acts as a reality check before intimacy can progress further, as people are less likely to misrepresent observable attributes in real-life settings (Finkel et al., 2012).

The goal of using these domains can vary. Daters looking for a temporary relationship often increase their expectations, whereas those looking to move the relationship permanently offline tend to decrease their expectations. In addition, daters who initiate communication are more likely to find suitable partners than those waiting to be asked (Kreager et al., 2014). Even though there are not many studies following daters long-term, the results from one large US national study (N = 19,131) demonstrated that one-third of marriages now begin online. It was also observed that those marriages which began online were less likely to result in divorce and were associated with slightly higher marital satisfaction than those that started in traditional offline venues. This data demonstrates that the internet alters the dynamics and outcomes of marriage (Cacioppo et al., 2013).

There are now multiple domains offering different types of relationship support. Apps such as Nayana are used to predict seven years of romantic outlook. The app predicts that someone who has remained in the same location for five years or is an introvert is less likely to meet people than someone who has just moved or is an extrovert. Another app, The Grade, gives grades to users based on their dating behaviour, giving them letter grades and suggestions for improvements. Those who ignore their advice get kicked out. There is a gap in research on online dating behaviour in different cultures, such as in India. Lastly, technological advances in online dating cannot replace the actual work required for an intimate relationship to advance (Wiederhold, 2015).

Study on Tinder

Tinder is currently the most popular dating app with certain advantages over traditional dating methods, such as enhanced portability of access (being an app) and the geolocalisation capacity (Schrock, 2015). Due to its high popularity, it enables increased access to potential matches. It is best in class for its simple and user-friendly interface design, with the most popular feature being the 'swipe' matching characteristic. It entices users by providing the stimulation of localised likes, making it easy for a potential meeting. However, compared to apps outside of the dating sphere, these features have an increased potential for a behavioural addiction due to this added element of romance associated with the use of an artificial domain (Orosz et al., 2016). The Uses and Gratifications Theory has shown the reason for the popularity of Tinder is because it fulfils physical (e.g. sexual desire), social (e.g. finding a partner or a friend) and psychosocial needs (e.g., validating one's physical attractiveness). Studies also suggest that tinder consumption may be used as a coping mechanism to regulate depressive effects and improve self-esteem (Orosz et al., 2018) with variable other motives depending on gender (Ranzini & Lutz, 2016).

The study consisted of 1,697 subjects, of whom 46.59% were females. All subjects completed all questionnaires and reported a heterosexual orientation and were thus retained in the final analysis. Subjects' age ranged from 18 to 74 years old ($M = 30.02$, $SD = 9.19$). In the study, four clusters were formed. Cluster 1, "regulated," was the largest group and showed low impulsivity and high self-esteem with a secure attachment, low depressive mood and problematic use. Cluster 2, "regulated with low desire," was the smallest group that exhibited low/medium impulsivity, low self-esteem, anxious attachment, low motives, more depressive mood but low problematic use. Cluster 3 was "unregulated and highly motivated" and exhibited: high level of motives to use Tinder and a high level of anxious attachment, a high level of urgency and sensation-seeking, and a moderate level of self-esteem. This cluster had the highest level of problematic use and the greater number of current matches whilst reporting an intermediate level of depressive mood (Rochat et al., 2019). Tinder use in Cluster 3 subjects was prompted by a combination of solid motivations (social, enhancement, and coping) and lacking self-control, especially urgency, attributed to weak inhibitory control and poor decision-making (Rochat et al., 2018).

They also have trouble overcoming the immediate gratification provided by the app, for example, the excitement associated with potential sexual activity and reinforcement resulting from frequent offline contacts, as indicated by their high sensation-seeking levels. Accordingly, users with diminished self-control seem to be at higher risk for developing problematic Tinder use, perhaps because of an interaction between poor self-control and strong motives incentivising app use. People who had low self-esteem and a

high level of avoidant attachment could be more inclined to use social networking sites as these online environments seem safer to express themselves or meet potential partners. Furthermore, the app can satisfy a need for social or sexual communication without spatial (e.g., proximity) and social (e.g., possibility to unilaterally swipe left on a potential match without consequences) constraints (Sumter et al., 2017). In addition, it has been demonstrated that this subgroup has low conscientiousness, and it has been observed that Tinder use has been associated with alleviating boredom (Orosz et al., 2018).

There were limitations to this study, such as its cross-sectionalism. Research is required to confirm the validity of the clusters longitudinally. In addition, the sample was self-selected, hence limiting generalisability. It also relied on self-reports creating various biases (e.g., social desirability and lack of insight). Also, the sample contained only heterosexual-identifying subjects hence cannot be applied to non-heterosexual participants (Rochat et al., 2019). Corroborating prior studies on internet gaming depression (Billieux et al., 2015), the observations show that the dysfunctional use of Tinder is heterogeneous and requires a wide variety of psychological variables that interact. Further studies are needed to disentangle the role of these variables in problem use growth, perpetuation, and recurrence and explain the very socially impactful appeal this and similar dating apps have (Rochat et al., 2019).

Dispositional factors predicting use of online dating sites and behaviours related to online dating

The use of online dating sites was investigated concerning many dispositional considerations. The theory was that people who scored higher on extraversion, neuroticism, self-esteem, and rejection sensitivity were more inclined to over-use a dating site. Specifically, rejection sensitivity was a significant factor in predicting the use of online dating sites, where personalities with higher rejection sensitivity were more likely to use a dating site (Blackhart et al., 2014).

This is suggested to be due to the ease of avoiding potential rejection in an online dating environment rather than in traditional dating environments. For example, online dating platforms provide better opportunities for self-representations as users tend only to present the pleasant side of themselves, therefore, reducing potential rejection. Furthermore, it is easier to manage communication when communicating online, and rejection online is less salient than face-to-face rejection (Blackhart et al., 2014).

The collective data offers some positive validation since traditionally, partnerships that have been formed online have been frowned upon due to societal stigma. With recent statistics showing cell phones becoming more popular and people's lives becoming more hectic than before, it is becoming a more appealing choice than meeting people

organically. As with previous studies, it is proposed in this paper that the use of such media can be healthy but has the potential to become harmful in the wrong hands, such as those with dysfunctional attachment issues. In addition, if also potentially struggling with specific mental health problems such as depression, a dating site can become a coping mechanism and therefore has the propensity to cause further long-term harm by deferring the real problem. Despite this, evidence indicates that a small group of individuals, and, more importantly, the use of such websites, will typically form a long-lasting and happy partnership.

Conclusion

It is clear from the research that social networking is now important for humanity, from running a business to finding a spouse to connecting with friends and family. The neuropsychology research shows how well adapted social media suits our evolutionary needs to keep up our reputation, observe and compare ourselves to other people and feed our desire to be liked. This is reflected in the supply of such features on all the most visited sites. The research shows that the combination of immediate gratification and relief from loneliness and young age and genetic factors may result in potentially harmful consequences such as dependence and addiction.

There are clear benefits of using social networking, primarily to offer worldwide social and emotional support to people. With knowledge of the possibility of substantial damage to more vulnerable individuals and awareness of the harmful functional properties of social media features, several cautionary steps in an ethical direction could provide a well-rounded, desirable platform.

To gain real social benefits and avoid the pitfalls of technology taking over genuine relationships, a network that allows initial connectivity but quickly leads to face-to-face contact would be an apparent alternative solution. In terms of amending what is already there, this could be achieved by several approaches, including a focus on social and emotional encouragement without the possibility of lurking other public accounts or providing hollow validations with strategies such as liking.

Since all similar research has been conducted on a small scale, this article contains data to justify a more extensive scale analysis into the usage and value of various social media facets.

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Appendix

<u>Negatives of social media use</u>	<u>Mechanisms of how social media works</u>	<u>Positives of social media use</u>	<u>Applications</u>
People reliant on validation – addition & high levels of anxiety linked to social media use	Dopamine release linked to giving likes & receiving likes & lurking	Social, emotional and self-esteem support	Providing social and emotional support without the option to give or receive likes & no personal information provided
Often misleading or inaccurate without any solid evidence	Anyone has access to make blogs/ spread information	Factual information worldwide availability of quick information spread	All information provided evidence-based
Misused due to the anonymity it provides, freedom to overuse	Providing platform/ forums for discussion	Allows communication despite spatial/ social and temporal distance	Time limits on usage and verified profiles whilst providing platforms
Misused often to social interaction face-to-face, an increase of anxiety due to fear of missing out, comparison to other people	The constant input of information about new and people/ continuously new content provided regarding everything and anything	Keeps occupied/ diminishes boredom/ but also can be used for education to learn something new or motivate	Education and communication part only
Anonymity/ reliance	Easy to access platforms that everyone has makes it easy to contact and arrange	Allows people to make plans/ arrange gatherings	Providing the platform without the possibility of anonymity and reliance by having verified accounts only and usage limits